THE DUAL OF A FINITE I-PROJECTIVE MODULE IS I-PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Definition of an I-projective module through annihilation of extension classes. introduced in the same section, source lines 17780-17810. Supplies the hypothesis on M and the form the conclusion for the dual has to take.
- A finite module admits a surjection from a finite free module whose kernel description gives the Ext class. used at proof lines 17834-17838 where the sequence 0 to K to R^r to M to 0 is chosen. Produces the presentation along which the factorisation property is dualised.
- Dualising a surjection of modules gives an injection, and dualising a factorisation produces a factorisation in the opposite direction. used at proof lines 17839-17843. Turns the factorisation of multiplication by a on M through R^r into the corresponding factorisation on the dual, which is the conclusion.

Source reference. Original lemma 0G96: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20291–20291. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. If M is a finite, I-projective R-module, then $M^\vee = \mathrm{Hom}_R(M, R)$ is I-projective.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Assume M is finite and I-projective. Choose a short exact sequence $0 \to K \to R^{\oplus r} \to M \to 0$. This produces an injection $M^\vee \to R^{\oplus r} = (R^{\oplus r})^\vee$. Since the extension class in $\mathrm{Ext}^1_R(M, K)$ corresponding to the short exact sequence is annihilated by I, we see that for any $a \in I$ we can find a map $M \to R^{\oplus r}$ such that the composition with the given map $R^{\oplus r} \to M$ is equal to $a : M \to M$. Taking duals we find that $a : M^\vee \to M^\vee$ factors through the map $M^\vee \to R^{\oplus r}$ given above and we conclude. $\square$